QUERRIEL ARVY MENDOZA

qasmendoza@gmail.com | LinkedIn | GitHub | Google Scholar

PhD student researching embedded AI and Micro-ROS | Seeking Summer 2027 robotics/embedded systems internship

EDUCATION

Ph.D. in Computer Science — Kansas State University, Manhattan, KS Expected 2027
GPA 3.9 / 4.0 | Research: Embedded systems, edge AI, and Micro-ROS on NVIDIA Jetson Orin and ESP32

M.S. in Computer Science — Kansas State University, Manhattan, KS 2023
GPA 3.8 / 4.0 | Thesis: Real-time deep learning deployment for insect detection on NVIDIA Jetson Nano

B.S. in Electronics Engineering — Adamson University, Manila, Philippines 2009
GPA 3.75 / 4.0

TECHNICAL SKILLS

Embedded & Robotics: NVIDIA Jetson Orin / Nano, ESP32, Micro-ROS, ROS 2, FreeRTOS, DDS / XRCE-DDS, Embedded Linux, I^2C, SPI, UART
Languages & AI: C, C++, Python, SQL | PyTorch, TensorFlow, TensorRT, ONNX, MobileNet-SSD, CNNs, Edge Inference, Computer Vision
Tools & Platforms: Git, GitHub, Linux, Docker, CMake, AWS, Azure, Azure DevOps, JetPack SDK, Power BI, Pandas, NumPy

RESEARCH EXPERIENCE

Graduate Research Assistant — Embedded AI & Robotics Dec 2023 – Present
Kansas State University, Department of Computer Science | Manhattan, KS

- Architecting heterogeneous robotics systems pairing NVIDIA Jetson Orin (perception/learning) with ESP32 (real-time control) via Micro-ROS over DDS-XRCE for edge-AI on resource-constrained hardware.
- Building real-time Micro-ROS publisher / subscriber nodes in C / C++ on ESP32 with FreeRTOS, characterizing end-to-end latency and reliability of sensor streaming to Jetson Orin.

Graduate Research Assistant — Machine Learning & Computer Vision Aug 2021 – Dec 2023
Kansas State University, Department of Computer Science | Manhattan, KS

- Built a real-time insect-detection vision pipeline reaching 85–90% accuracy and 81% F1 across lighting conditions by training MobileNet-SSD on 1,000 annotated images in PyTorch and deploying via ONNX → TensorRT on NVIDIA Jetson Nano 4 GB.

PROFESSIONAL EXPERIENCE

Data Science Intern — Caterpillar Inc., Chicago, IL May – Aug 2023

- Generated ~$2 M in projected cost savings by building and deploying ML models in Python / SQL to predict asset and event criticality, surfaced via Power BI for senior stakeholders.
- Surfaced decision-ready insights from large operational datasets while collaborating with a cross-functional team in an agile Azure DevOps workflow.

IT Operations Lead — Accenture, Manila, Philippines Oct 2015 – Oct 2016

- Improved client system performance by 30% (Value Creator Award) by designing and deploying enterprise storage and backup solutions.
- Sustained high availability and SLA/KPI compliance by leading and supervising a 30+ person multi-platform IT operations team.

Infrastructure Engineer (Storage & Backup) — Ingram Micro, Manila, Philippines Jul 2014 – Oct 2015

- Ensured reliable data protection for newly provisioned servers by designing and administering storage / backup configurations across EMC, DataCore, and NetApp systems; completed AWS training.

Service Delivery Consultant (Storage & Backup) — HP Enterprise, Manila, Philippines May 2010 – Jul 2014

- Increased system productivity by 25% (Received Gold Award) by designing and implementing enterprise storage and backup solutions for newly built servers.
- Reduced backlogged tickets by 45% (Received High Profile Award) by executing critical platform-migration work and improving backup success via Problem / Change Management.

PUBLICATIONS

- Serfa Juan, R. O., **Mendoza, Q. A.**, Neilsen, M., & Gerken, A. (2026). “AI-Driven Morphometric Fragment Analysis for Automated Post-Harvest Identification of Tenebrionidae Species.” *IEEE Green Technologies Conf. (GreenTech)*, 470–475. DOI: 10.1109/GreenTech68285.2026.11471609
- **Mendoza, Q. A.**, Pordesimo, L., Neilsen, M., Armstrong, P., Campbell, J., & Mendoza, P. T. (2023). “Application of Machine Learning for Insect Monitoring in Grain Facilities.” *AI (MDPI)*, 4(1), 348–360. DOI: 10.3390/ai4010017
- **Mendoza, Q. A.**, et al. (2023). “Enhancing Grain Facility Management with an AI-Based Insect Detection and Identification System.” *SPIE Defense + Commercial Sensing*, Paper 12545-33. DOI: 10.1117/12.2672253
- **Mendoza, Q. A.** (2023). “Integrating an AI-Based Insect Detection and Identification System to Optimize Grain Facility Management.” *M.S. Thesis, Kansas State University*. hdl.handle.net/2097/43529

CERTIFICATIONS & AWARDS

Machine Learning Specialization (DeepLearning.AI) • DevOps on AWS (AWS) • Data Analysis & Visualization (IBM) • Value Creator Award (Accenture) • Gold Award (HP Enterprise)